

Bradley Brewington

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EDUCATION

Texas A&M University — B.S. Mechatronics Engineering Technology

College Station, TX | May 2026

Coursework: Mechatronics I & II, Robotic Systems Design, Controls, Embedded Systems, Analog Electronics, Electronic Instrumentation, Mechanical Design Applications, Strength of Materials

TECHNICAL SKILLS

Mechanical: SOLIDWORKS (parts, assemblies, dimensioned drawings, Flow Simulation), Autodesk Fusion 360, FDM 3D printing (Creality K1C), shop-floor rework (bandsaw, drill press, rotary tool)

Electrical / Hardware: Multi-layer mixed-signal PCB design (KiCad), schematic capture, SMT/THT assembly, hot-air rework, contract-fab release (Gerbers/BOM/CPL), LTSpice

Embedded / Controls: Bare-metal MCU (TI MSP432/MSP430, Arduino, ESP32), ARM Assembly, ADC/PWM/GPIO/UART/I2C/SPI, stepper control (BTT SKR / Marlin / NEMA17 / TMC2209), PID tuning (coursework), MATLAB/Simulink

Robotics / Software: ROS, OpenCV, LIDAR, ToF, sensor fusion, state-machine autonomy; Python, C/C++ (coursework), Git/GitHub, Linux

Bench / Debug: Oscilloscopes, signal/spectrum analyzers, function generators, programmable supplies, multimeters; structured fault isolation, root-cause investigation

PROFESSIONAL EXPERIENCE

Research Assistant, Hardware Engineering | Baker Research Group, Texas A&M

April 2024 – May 2026

- Built 4-axis stepper motion system for an electrochemical-experiment cell-imaging rig: BTT SKR Mini E3 V2.0 controller, 4x NEMA17 steppers with TMC2209 drivers, wiring harness, Marlin firmware over USB serial. Operational and in use.
- Diagnosed motor instability presenting as firmware corruption; M122 status report confirmed driver UART dropout on each move.
- Traced root cause to stepper cables splitting each motor coil across both driver pairs; re-pinned connectors against multimeter-verified coil pairs to fix the platform.
- Authored handoff operating manual: G-code reference, M122 fault-signature table, coil-wiring verification procedure.
- Designed 4-layer mixed-signal precision impedance PCB under senior technical direction (full schematic and layout in KiCad, contract-fab release); led ~6-month bench debug of a high-frequency artifact, isolating it point-by-point against a known-good reference and cross-checking in LTSpice.

Electronics Engineering Intern | InTech Defense, Chantilly, VA

May – August 2023

- Built customer-deliverable electromechanical hardware on DoD-contract programs: SMT/THT assembly, hot-air rework, solder reflow, harnessing, shielding-subassembly integration.
- Performed shop-floor mechanical rework (bandsaw, drill press, rotary tool) to resize parts and add mounting features on filter and shielding subassemblies.
- Supported the lead engineer on root-cause troubleshooting of new product variants and spectrum-analyzer-based RF shielding-effectiveness testing against contract spec.

PROJECTS

Portable Spirometer — Senior Capstone | SOLIDWORKS CFD, Fusion 360, FDM, BLE (team of 6)

- Mechanical and integration lead on a handheld BLE-connected pulmonary measurement device; owned the mechanical-to-electrical interface and coordinated PCB form factor with the electronics teammate.
- Designed full Fusion 360 assembly (mouthpiece, sensor PCB housing, base enclosure) with dimensioned drawings, iterated V1 → V2; ran SOLIDWORKS Flow Simulation across 0.5–14 L/s clinical flow rates; printed all parts in-house on a Creality K1C across multiple build cycles.

Autonomous Mobile Robot — MXET 300 | Python, ROS, LIDAR, OpenCV (team of 4)

- Owned target tracking and patrol-mode navigation on an autonomous mobile robot: fused LIDAR distance with OpenCV color-thresholded vision into a state-machine controller driving closed-loop PWM motors. Separately contributed mechanical arm design (4-bar linkage, 4.167:1 gear ratio) on a MXET 250 humanoid-hand robot.

Bioinspired Robotics — Peristaltic Worm & RC Animatronic Turtle | Arduino, servos, steppers (personal)

- Built a peristaltic-locomotion worm with Arduino-sequenced servos and molded liquid-latex setae anchors (climbed 60° inclines); reverse-engineered an RC car control board to drive Arduino-controlled hybrid actuation on an animatronic turtle.

LEADERSHIP

- Texas A&M Corps of Cadets, Ross Volunteer Company — Training Sergeant (2023–2025).** Member of Texas' oldest student honor guard (est. 1887) and honor guard to the Governor of Texas; ran training and accountability for 60+ cadets under hierarchical military standards.